1 Scope

The question is whether every threat model in the multi-party learning literature can be written with one set of objects: datasets, learning functions, a hypothesis class, and the models in it. The adversary is then fixed by what it sees, what it chooses, and one of three goals, with no named attack families. The method is to define the interface, map the papers placed on 2026-10-16 onto it, and instantiate it on HE-OFT (KSK26) and on FedAvg, in the terms and symbols of HE-OFT.

2 Result

Every adversary that the sweeps report is a triple: a view of the outputs of the protocol's learning functions, a set of protocol inputs it chooses, and a goal that is one of distinguish, invert or steer. Membership inference is a discriminator, reconstruction and model extraction are function inversion, and poisoning and backdoors are steering. Proven: the strength order between two adversaries is post-processing of views together with inclusion of choices, so black-box against white-box is a question of sufficient statistics. HE-OFT already proves one ordering in this language, that its label view is no stronger than holding the shared head, and it proves that a malicious server must be excluded. What HE-OFT leaves unclaimed is steering by deviating clients, and the check that would let it admit a deviating server.

Does not show: The interface is a definition, and definitions in this area are commodities, so its worth lies only in the collapse and separation results it organizes. Whether its novelty survives is not checked. Salem et al., IEEE S&P 2023, unify inference attacks as privacy games (from memory, not placed), and the distinguish and invert half may already be theirs. Every note except HE-OFT's is built from a sweep extraction, so no other row is cited from a full reading.

3 How it was produced

Sources	Four sweeps, venue lists of IEEE S&P, USENIX Security, CCS, NDSS, PoPETs, NeurIPS, ICML, ICLR, CRYPTO, EUROCRYPT. DBLP was rate limited. HE-OFT read in full from `HE-IFD/docs/paper`, commit 9d894ea.
Range	2021 to 2026, anchors before 2024 only where they fixed a threat model
Screened	44 papers placed with first-page author checks
Artifacts	`lib/literature/catalog.md`, `lib/literature/notes/KSK26.md`, `My Drive/HE-ML/pdfs/`
Reproduce	`lib/literature/positioning.md`

4 What this changes

A research problem in this language is one step between two adversaries that the field crosses without a result, with a collapse or a separation proven at a fixed goal and a test suite that measures the gap. For the group, the semi-honest to malicious server step is closed by HE-OFT's own impossibility, so the open object is the cheapest check on what the quorum switches that restores the semi-honest view, which is the question of the verifiable HE study. Küpçü's black-box question asks for the reverse of HE-OFT's ordering, when Q label queries reach what the shared head reveals, and this is open.

5 Method

Objects

Let $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$ be the data space. A dataset $D \in \mathcal{Z}^*$ is a finite multiset. Client $j \in \{1, \dots, N\}$ holds D_j of size n_j, and D_{pre} is the pretraining dataset. Let $\Theta \subseteq \mathbb{R}^p$ be the parameter space and $\mathcal{H} = \{h_\theta : \theta \in \Theta\}$, $h_\theta : \mathcal{X} \to \mathcal{Y}$, the hypothesis class. A model is one h_θ.

A *learning function* is a randomized map $\mathsf{A} : \mathcal{Z}^* \times \Theta \to \Theta$, where $\mathsf{A}(D; \theta)$ is the model obtained by

training on D from θ. Pretraining, K steps of SGD, LoRA with a frozen down-projection, and head training are all learning functions. The update map is $\mathsf{U}(D;\theta) = \mathsf{A}(D;\theta) - \theta$. For one gradient step $\mathsf{U}(D;\theta) = -\eta\nabla_\theta \ell(D;\theta)$, the derivative of the dataset's loss evaluated at the point θ. A *merge* is a map $\mathsf{M} : \Theta^N \to \Theta$.

A protocol is a finite directed acyclic graph whose nodes apply learning functions, merges and the query map $x \mapsto h_\theta(x)$. Every node value v is a function of $(D_{\text{pre}}, D_1, \ldots, D_N)$, the honest randomness, and the adversary's choices.

The adversary

Each node value reaches the adversary through a channel $\kappa_v \in \{\text{id}, \Sigma, \mathsf{Enc}, \mathcal{N}_\sigma, \perp\}$: the plaintext value, a sum over a set of clients, a ciphertext under a key the adversary does not hold, added noise, or nothing. A decryption policy releases $g(v)$ for some functions g of encrypted values, and for approximate HE the released value carries the decryption noise as well. The view is

$$\mathsf{View} = \big(\kappa_v(v)\big)_{v\in S} \text{ together with oracle access to } x \mapsto o(h_\theta(x)),$$

for the set S of node values it sees and an output map o (identity for scores, arg max for labels, the loss for loss access).

The choices C are the protocol inputs the adversary sets: its own dataset, its own update, the point θ at which honest clients evaluate U, the merge M, the learning function A that clients run, query points, and the ciphertexts presented for decryption. A semi-honest adversary chooses nothing but query points. Every malicious model in the placed papers chooses more.

The goal is one of three games, each with a baseline, the best value reached from public knowledge alone.

- Distinguish. Two hypotheses H_0, H_1 on $(D_1, \ldots, D_N)$. The discriminator outputs a bit, and $\mathsf{Adv} = \Pr[1 \mid H_1] - \Pr[1 \mid H_0]$. Membership is a pair that differs in one record of D_j. Property inference is a pair of distributions for D_j. Telling a pretraining member from a fine-tuning member is $z \in D_{\text{pre}}$ against $z \in D_j$. HE-OFT's content game is this goal with $D_j \in \{D_0, D_1\}$ of equal size.
- Invert. A target τ of the datasets or the model, and a tolerance η. The adversary wins when $d(\hat{\tau}, \tau) \le \eta$. Reconstruction takes τ a record of D_j, and model extraction takes $\tau = \theta^*$ from the query oracle, scored by fidelity.
- Steer. A target set $T \subseteq \Theta$. The adversary wins when the trained model lands in T under its choices, while a defender's discriminator between honest and adversarial updates has advantage at most δ. Untargeted poisoning takes T the models of high risk, and a backdoor takes T the models that send a trigger set to a fixed label.

The stealth constraint on steering is itself a distinguish game, played by the defender. A client that tests whether the server deviated plays the same game.

The order, and the two kinds of result

Write $(\mathsf{View}_1, C_1) \preceq (\mathsf{View}_2, C_2)$ when $C_1 \subseteq C_2$ and $\mathsf{View}_1 = g(\mathsf{View}_2, r)$ for a randomized map g with independent randomness r. Proven, by the data-processing inequality for distinguishing and by simulation for inversion: $\preceq$ implies that the first adversary's best distinguishing advantage and best inversion success are at most the second's. The best distinguishing advantage equals the total variation distance between the laws of View under H_0 and H_1 (Neyman and Pearson). For efficient adversaries, g must be efficient.

A *collapse* at a goal shows that the weaker adversary attains the stronger one's best value. For distinguishing this holds exactly when View_1 is a sufficient statistic of View_2 for (H_0, H_1). A *separation* is a dataset law, learning function and hypothesis class with a large gap. Instances, as the sweeps report them unless marked:

- Labels against the head, distinguish. Ordering proven in HE-OFT, read in full: $Q_{\text{tot}} = (t-1)Q$ label answers give a deviating coalition no more than both shared heads in plaintext, and the white-box advantage measures near zero (KSK26).
- Loss against parameters, distinguish. Collapse under a Gibbs law on the parameters (Sablayrolles et al., ICML 2019, from memory). Separation claimed for SGD (Suri, Zhang and Evans, TMLR 2024, outside the venue rule).
- Each client's update against the sum, invert. Collapse when the server chooses a different θ for each client, since Σ then equals one update (Pasquini, Francati and Ateniese, CCS 2022).
- Honest against chosen θ, invert. Separation by construction, proven under an RIP condition (ARES, Gong et al., S&P 2026) and empirical (SEER, Garov et al., ICLR 2024, and LOKI, Zhao et al., S&P 2024).
- Chosen θ through a local DP channel, distinguish. Proven lower bounds for every LDP mechanism (Nguyen et al., ICML 2025).
- Steer with one's dataset against steer with one's update. Proven separation in stability for convex losses (Boudou et al., ICML 2026).
- Steer against the defender's discriminator. Proven for distance-based aggregators that perfect detection implies no learning (Raynal and Troncoso, S&P 2025).

HE-OFT on the interface

The pretraining dataset gives the backbone φ, public and frozen. Client j runs one learning function on D_j from φ and the public head initializer θ_0, and obtains its adapter A_j, kept local, and its head h_j. Its update is $\mathsf{Enc}(\mathrm{diag}(g_j)\Delta_j)$ and $\mathsf{Enc}(g_j)$, with $\Delta_j = h_j - \theta_0$ and g_j the per-class counts. The merge is the coverage-weighted average. Row c of the shared head is θ_0 plus $\sum_j n_{j,c}\Delta_{j,c} / \sum_j n_{j,c}$, evaluated under encryption. The shared head θ^* is never decrypted. A querier encrypts its own features, the server applies θ^* and an encrypted argmax, masks every slot but the label, and a quorum of t clients key-switches the label to the querier, up to Q times per client.

- Semi-honest server with $t-1$ clients. The view is ciphertexts, the public counts n_j, the selected index and the coalition's own answers. Proven simulation security under IND-CPA and a smudging hypothesis. The paper reports that its measured configuration does not meet that hypothesis and recommends 2^{25} smudging with a duplicate-query check (KSK26).
- Deviating clients, honest server. The choices are the clients' uploads, shares and arbitrary feature vectors as queries. The distinguish goal is bounded by a negligible term plus what Q_{tot} label queries give, under three readings on deviating messages that HE-OFT does not enforce by proofs. No steering claim is made. Proven, from the merge formula: since no check ties g_j to n_j or to the data, a client that reports a large $g_{j,c}$ moves row c of θ^* as close to its own $\Delta_{j,c}$ as it likes.
- Deviating server. Its choices include the ciphertext presented to the quorum. HE-OFT proves that a client's decision to contribute a share cannot depend on the plaintext, so no protocol in which clients decrypt what the server presents admits a deviating server, and at four classes one request carries the whole head (KSK26). The paper names a checker that recomputes the deterministic serving circuit, or verifiable HE, as the remedy.

FedAvg, split and vertical learning on the interface

In FedAvg, A is K local SGD steps and the server sees $\mathsf{U}(D_j;\theta)$ at honest points, or $\Sigma_j\,\mathsf{U}(D_j;\theta)$ under secure aggregation. Gradient inversion is the invert game on $D_j \mapsto \mathsf{U}(D_j;\theta)$. A malicious server chooses θ to make that map injective with a computable inverse (ARES and LOKI), or

chooses it per client to remove Σ (Pasquini, Francati and Ateniese). With LoRA, U is confined to the adapter, and it stays invertible from the low-rank gradient while the token count is below the embedding dimension (DAGER, Petrov et al., NeurIPS 2024). In split learning the view is the client's features and their gradient, and a malicious server chooses the loss that trains the client's layers, so its choices contain A itself (Pasquini, Ateniese and Bernaschi, CCS 2021). In vertical learning the label holder chooses the gradients it returns, and the URVFL attack uses this to invert the other parties' features (Yao et al., NDSS 2025).